

Proactive Memory for Event-Driven LLM Agents

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Abstract

Proactive LLM agents anticipate service opportunities from ongoing observations and accumulated history. A central challenge is determining which historical observations remain applicable to the current event and can support a timely service decision. We propose Proactive Memory (ProactMem), combining bounded recent events, source-checked temporal trajectories, and replay-assessed behavioral regularities. The trajectory validation step checks extracted facts against source events, while historical replay assesses recurring regularities before they are matched to current events. We also construct ProactiveStream, a longitudinal benchmark covering three application scopes, where agents process isolated event streams chronologically under episode-aware opportunity labels. ProactMem achieves higher effectiveness than the evaluated external baselines in every scope. Ablation studies support the contributions of hierarchical memory organization and validation to service-opportunity detection.

1 Introduction

Proactive LLM agents anticipate useful assistance as a user’s situation evolves, taking initiative before an explicit request. Recent systems study when to propose a task, detect emerging needs, and use contextual or historical information to guide that judgment (Lu et al., 2025; Yang et al., 2025; Xie et al., 2026; Li et al., 2026). Across our Sports, Home, and Code scopes, this requires interpreting a continuing stream of observations and deciding whether a service opportunity warrants attention now.

Historical context is central to this decision. Figure 1 illustrates how earlier evening walks can inform a service judgment when current activity is low and the user is available. The temporal trajectories and behavioral regularities retain this context beyond a recent-event window. The current event

guides regularity matching, and retrieved memory informs the service decision.

Agent memory systems retain observations, consolidate experience, and retrieve related information (Park et al., 2023; Packer et al., 2023; Chhikara et al., 2025; Xu et al., 2025). To turn this history into decision context, proactive agents must assess whether an abstraction reflects its source events, whether a proposed behavioral regularity has recurring support, and whether that regularity applies now. Our central question is *how to turn longitudinal history into reliable, currently applicable context for proactive service decisions*.

We propose Proactive Memory (ProactMem; Figure 2), a hierarchical memory framework that organizes event histories into source-checked temporal trajectories and replay-assessed behavioral regularities for proactive service decisions. The bounded event memory preserves recent observations, while temporal trajectory memory retains source-checked abstractions beyond the recent window. The behavioral regularity memory consolidates recurring conditions from these temporal trajectories and assesses their historical support through replay. At decision time, the agent combines recent events and temporal trajectories with currently applicable regularities. The decision LLM assesses whether current observations or historical context support a concrete, timely, low-risk service; intervention control tracks whether the proposed opportunity has already been covered.

We construct *ProactiveStream* to study these decisions over isolated longitudinal event streams. Its three scopes preserve causal history and define episode-aware reference opportunities, allowing evaluation of memory formation and repeated service decisions within the same streams.

Our contributions are:

- **Method.** ProactMem combines source-checked trajectory abstraction, replay-based

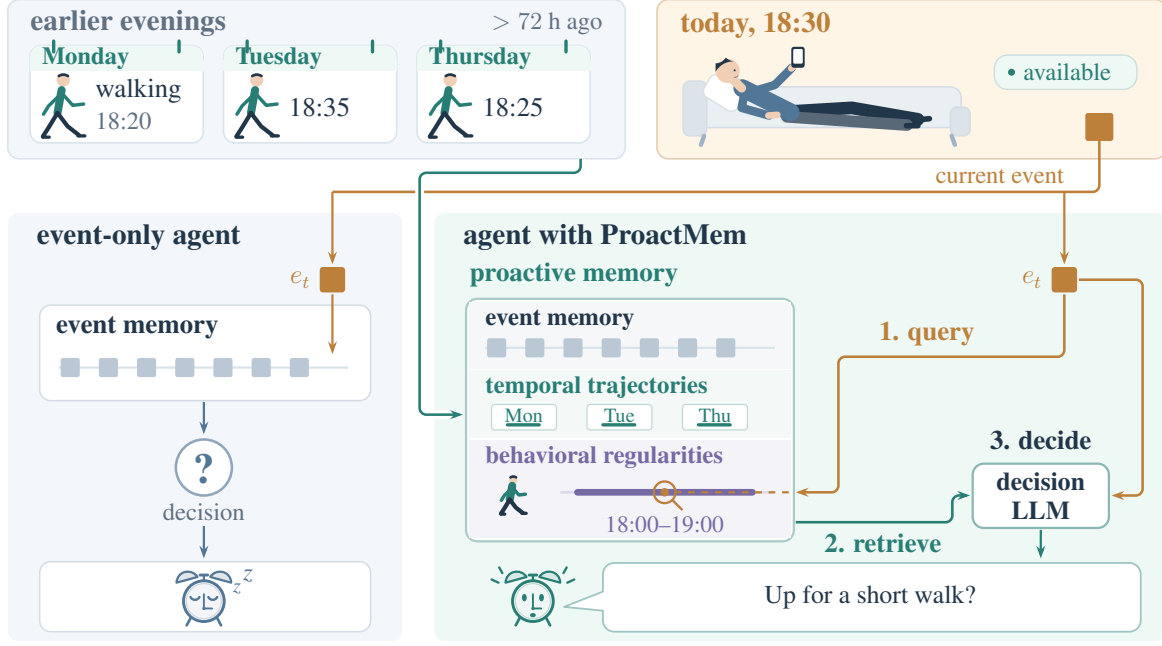


Figure 1: Illustrative Sports scenario. Both agents receive the same current event and recent history. The current event queries ProactMem to match stored behavioral regularities. Retrieved regularities join recent events and temporal trajectories in the decision context. Responses depict possible decisions.

regularity assessment, and selection of currently applicable regularities to supply historical context for proactive service decisions.

- **Benchmark.** ProactiveStream supports longitudinal evaluation of memory formation and repeated service decisions, preserving causal history, entity isolation, and episode-aware reference opportunities across three scopes.
- **Empirical findings.** ProactMem achieves higher event-level effectiveness and balance than the evaluated external baselines in all three scopes; targeted ablations support the contributions of hierarchical organization and validation.

2 Problem formulation

Let u denote an independently evaluated entity, such as a participant, household, or repository, with a time-ordered public event stream

$$E^u = (e_1^u, \dots, e_{T_u}^u), \quad e_t^u = (\tau_t^u, x_t^u), \quad (1)$$

where T_u is the number of events for entity u , τ_t^u is the event time, and x_t^u contains the scope-specific public fields observable at that time. We omit the entity superscript u when it is unambiguous.

Events follow a fixed chronological order with deterministic tie-breaking. Before processing e_t^u , an agent has a causal state Z_t^u constructed only from events and finalized outputs preceding e_t^u in this order. It produces

$$(\hat{y}_t, \hat{m}_t) = \pi(e_t^u, Z_t^u), \quad \hat{y}_t \in \{0, 1\}, \quad (2)$$

where $\hat{y}_t = 1$ denotes *intervene*, $\hat{y}_t = 0$ denotes *abstain*, and $\hat{m}_t$ is an optional evidence-grounded service message. Here π denotes a general inference policy over the current event and causal state; $\hat{m}_t$ may be empty under *abstain*. After finalizing the decision and optional message, the agent updates its state through an update function U :

$$Z_{t+1}^u = U(Z_t^u, e_t^u, \hat{y}_t, \hat{m}_t). \quad (3)$$

Each entity maintains isolated state, and π accesses only the current public event and causally available same-entity information.

The objective is to identify a timely, actionable *service opportunity* supported by the current event or prior same-entity observations. Let $y_t \in \{0, 1\}$ denote the frozen reference decision for e_t under the annotation protocol in Section 4.2; all evaluated policies share these fixed references. Labels and annotation metadata remain evaluator-private. Events with either reference label share the same scope-specific public input schema.

3 Proactive Memory (ProactMem)

ProactMem uses three complementary memory layers (Figure 2): event memory for recent public events, temporal trajectory memory for source-checked activity summaries, and behavioral regularity memory for replay-assessed conditional regularities. Construction checks source fidelity and historical support, retrieval selects currently applicable regularities, and the decision LLM assesses present service value.

3.1 Memory construction and update

Service opportunities can depend on both recent changes and recurring behavior, motivating a memory hierarchy that preserves fine-grained evidence while validating longer-term abstractions.

bounded event memory. Evidence utilization can degrade in long contexts (Liu et al., 2024), motivating a bounded event layer that preserves recent observations at their original granularity. For event e_t , event memory retains prior events in their original public representation, intersecting a temporal horizon Δ with capacity K :

$$S_t = \text{Tail}_{K-1}((e_i)_{i < t, 0 \leq \tau_t - \tau_i \leq \Delta}). \quad (4)$$

Here Tail_{K-1} selects the most recent $K-1$ events in the filtered sequence. We use $\Delta = 72$ hours and $K = 100$: at most 99 prior events accompany the separately supplied current event. An independent daily buffer retains all committed events of the open local day, preserving complete input for trajectory construction.

fact-validated temporal trajectory memory. To retain activity context beyond the recent window while keeping abstractions traceable, we construct temporal trajectories with source-level fact validation. After a local day closes, a construction LLM summarizes its buffered events as trajectory segments with intervals, typed facts, and event citations. Let $F(\cdot)$ denote normalized checkable facts obtained from typed fields. For segment d and its cited events $\mathcal{E}(d)$, source consistency requires

$$F(d) \subseteq \bigcup_{e \in \mathcal{E}(d)} F(e), \quad \mathcal{E}(d) \subseteq E_{\text{completed day}}^u. \quad (5)$$

Here $E_{\text{completed day}}^u$ contains the same entity’s events from that completed day. The validator checks entity, interval, citations, and canonicalized scalar

values at compatible field paths. Acceptance requires source-supported typed facts and resolved citations. Accepted temporal trajectories form W_t , retaining historical context beyond event memory. Appendix B details field handling and buffering.

replay-assessed behavioral regularity memory. To ground behavioral expectations in observed recurrence, we assess proposed conditional regularities through historical replay and revise their eligibility as evidence accumulates. Weekly consolidation groups seven completed daily records per entity in time order, with each record assigned to one cycle. A cycle may span more than seven calendar days. An LLM proposes a trigger g , expected behavior p , time window, optional weekday restrictions, and source daily identifiers. Replay checks these conditions against validated daily facts: an observable triggered day is fulfilled when p occurs at or after the trigger within the window and violated otherwise. Missing observations are censored, inapplicable days are excluded, and each date contributes at most once. Let n_f count fulfilled (supporting) days and n_v count violated days:

$$\text{conf}_{\text{replay}} = \frac{n_f}{n_f + n_v}, \quad \text{lift} = \frac{\text{conf}_{\text{replay}}}{\text{base}(p)}. \quad (6)$$

The base rate is the frequency of p over observable days under the same time-window and weekday restrictions across both triggered and other observable days. Zero denominators yield zero confidence or lift, respectively.

A well-formed proposal becomes *active* when $n_f \geq 3$, confidence ≥ 0.5 , and lift ≥ 1.0 ; otherwise it remains a decision-invisible *candidate*. Malformed proposals are rejected. Consolidation resets the online trajectory view, retaining records for audit. Later observations update regularity confidence and lifecycle: repeated violations can lower retrieval priority or suspend eligibility. Appendix B.2 specifies these transitions. Historical support characterizes observed recurrence; future applicability and present service value require separate assessment.

3.2 Memory retrieval and selection

To make accumulated regularities relevant to the present service opportunity, retrieval checks their trigger conditions, temporal applicability, and occurrence state. The event memory supplies the bounded sequence and W_t supplies all completed temporal trajectories in the current cycle. Retrieval

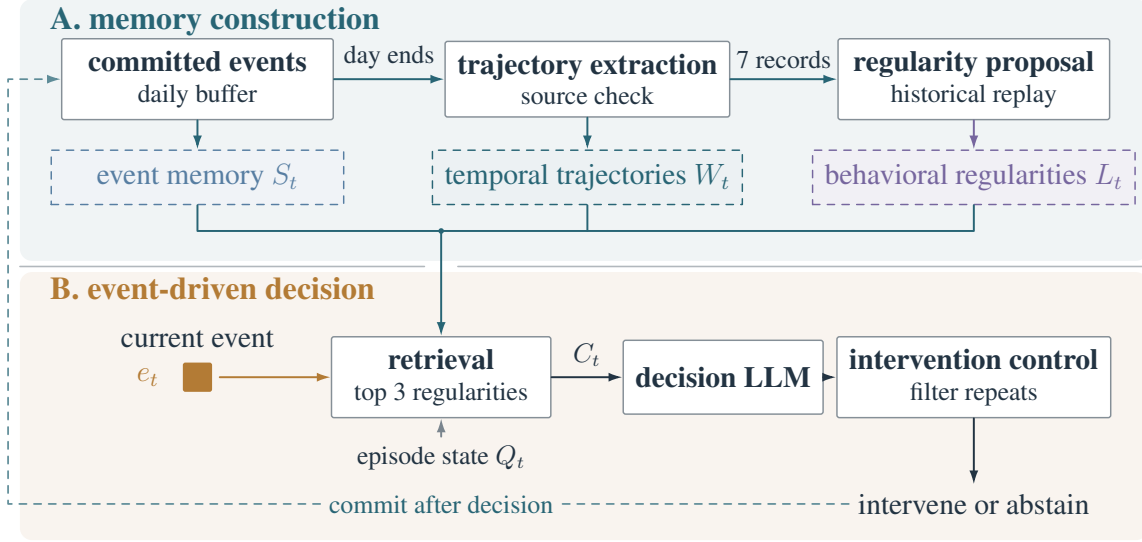


Figure 2: ProactMem architecture. (A) Completed history supplies bounded events; construction LLMs produce source-checked temporal trajectories and replay-assessed behavioral regularities. (B) The current event guides retrieval of up to three applicable regularities. Context C_t combines these regularities with recent events, temporal trajectories, the current event, and episode state Q_t . The decision LLM proposes assistance, intervention control filters repetition, and dashed feedback commits the event after finalization.

of regularities additionally checks entity-local trigger context, weekdays, and temporal applicability with a 30-minute lead and grace interval. Matching considers current and previously recorded causal trigger occurrences. An occurrence is fulfilled when its expected behavior has been observed; censored occurrences have insufficient observations for assessment. Selection excludes both states. For regularity store L_t^u , lifecycle eligibility Eligible_t , trigger and time matching Match_t , and occurrence state s_t ,

$$R_t = \text{Top3} \left\{ \ell \in L_t^u : \begin{aligned} &\text{Eligible}_t(\ell) \wedge \text{Match}_t(\ell), \\ &s_t(\ell) \notin \{\text{fulfilled}, \text{censored}\} \end{aligned} \right\}. \quad (7)$$

The top-three selection ranks eligible regularities by occurrence state, lifecycle, confidence, and temporal distance (Appendix B.2). All matching uses causally available observations to assess present applicability.

3.3 Proactive decision

Timely assistance requires judging current needs alongside historical context (Yang et al., 2025), so our decision stage assesses service value and tracks prior coverage of the opportunity. The decision context combines the current event, recent events, temporal trajectories, selected regularities, and ob-

servable episode state:

$$C_t = (e_t, S_t, W_t, R_t, Q_t), \quad (8)$$

where Q_t describes the service opportunity’s observable state; the controller separately records previously allowed interventions to track service coverage. The LLM produces $(\hat{y}_t, \hat{m}_t) = \pi_\theta(C_t)$, proposing intervention when the current event *or* historical context supports concrete, timely, feasible, low-risk assistance, and abstention otherwise. The LLM combines matched regularities with current context and can select either intervention or abstention.

Within each scope, all methods apply the same intervention controller policy to their own context and intervention history. The controller compares proposed services with previously allowed interventions and converts duplicates to *abstain*. Scope-specific changes in opportunity state, service identity, contextual regularities, or local day permit renewed assessment; each intervention originates from an LLM proposal. The controller finalizes the proposal as $(\hat{y}_t, \hat{m}_t)$, the output defined in Section 2. Appendix B.3 specifies matching fields, phase transitions, and re-entry rules.

Inference prepares prior memory, stages e_t , selects relevant context, obtains and finalizes the decision, and then commits e_t . Maintenance incorporating e_t can affect only later decisions. Ref-

erence labels remain evaluator-private throughout construction and retrieval.

4 ProactiveStream benchmark

ProactiveStream evaluates service-opportunity detection in longitudinal event streams, where agents assess incoming observations using their own accumulated history. Table 1 summarizes its composition and provenance across Sports, Home, and Code.

4.1 Scopes and public events

Sports uses participant activity and availability contexts for optional walking and everyday physical-activity assistance.

Home uses time-bounded home-sensor observations for gentle, low-risk check-ins. Each sensor segment becomes available at interval closure.

Code uses repository activity at GitHub’s native event granularity for low-risk triage, review, and maintenance, with one decision per event.

Each participant, home, or repository defines an isolated chronological stream, allowing earlier observations to inform later service opportunities. Public inputs contain contemporaneously available observations, using the same scope-specific schema for both label classes. Agents construct their own memory from these events; annotation and scoring metadata remain evaluator-private. Appendix A details source datasets, fields, and preprocessing.

4.2 Reference labels and causal replay

Annotators assess e_t and its complete prior same-stream history $\mathcal{H}_{<t}$. Let A indicate a concrete, timely, feasible, low-risk opportunity supported by e_t , B an opportunity at e_t supported by historical observations, and N_t^{ref} reference-episode novelty:

$$y_t = 1 \iff (A(e_t) = 1 \vee B(e_t, \mathcal{H}_{<t}) = 1) \wedge N_t^{\text{ref}} = 1. \quad (9)$$

The reference protocol labels the first reliable actionable window *intervene* and covered windows *abstain*, allowing renewed opportunities when support strengthens, the service changes, or an episode reopens. Reference novelty follows the reference opportunity sequence independently of model outputs and is shared across methods. The event-level labels therefore assess timely identification of service windows defined by the annotation protocol.

LLM-assisted screening prioritizes candidate opportunities; three team members then indepen-

dently label every event while blinded to evaluated-system outputs, followed by cross-review and adjudication. For a separate-team review, a fixed SHA256 rule selects one stream and 150 contiguous targets per scope from public metadata, requiring at least seven days of prior history and retaining each complete causal prefix. Three reviewers label these 450 targets with model outputs hidden and form consensus references. Section 5.2 reports reference-sensitivity scoring of fixed diagnostic outputs. Main comparisons use protocol-fixed evaluation on the jointly developed benchmark, with labels, prompts, memory bounds, and validation settings frozen before the corresponding runs.

Each stream is replayed chronologically from empty method state, with access to the current event and strictly prior same-stream context. Each decision is finalized before the current event is committed to memory; reference labels are joined after inference for scoring. Scoring includes the initial events and continues as history accumulates.

5 Experimental evaluation

We examine overall effectiveness (**RQ1**), memory organization (**RQ2**), and validation mechanisms (**RQ3**), with supplementary checks of reference sensitivity, episode behavior, and model choice.

5.1 Experimental settings

Baselines. We evaluate five external methods:

- **ProactiveAgent** uses its released prompt and proposal-or-null output contract, with causal history of events and decisions capped at 1,000 events (Lu et al., 2025; THUNLP, 2026).
- **ContextAgent** adapts in-context proactive-service judgment to public events (Yang et al., 2025).
- **Mem0** extracts, consolidates, and retrieves same-entity memories (Chhikara et al., 2025; Mem0, 2026).
- **A-MEM** constructs linked notes, updates associations, and retrieves relevant notes (Xu et al., 2025).
- **Graphiti** stores event relations in a temporal graph and retrieves same-stream facts (Zep, 2026).

Metrics. Scores use one post-controller decision per event; immediate and scheduled services count as *intervene* at the decision event. We report **relevance** (precision) as the share of interventions matching reference opportunities; **coverage** (recall)

Table 1: Composition and provenance of ProactiveStream across Sports, Home, and Code. Each stream is an independent memory and replay unit. Reference interventions denote service opportunities under the annotation protocol.

Scope	Primary source	Entity	Streams	Input events	Temporal coverage	Reference interventions
Sports	HeartSteps	participant	9	1,456	367 observed participant-days	365
Home	CASAS	home	4	1,624	14 consecutive days/home	190
Code	GH Archive	repository	4	1,459	longitudinal repository histories	174
Total	three scopes	isolated stream	17	4,539	three longitudinal settings	729

is the share of opportunities recovered. Their harmonic mean, **effectiveness** (F1), balances relevant interventions against missed opportunities. The **intrusion** score (false positive rate) is the share of reference negatives receiving interventions. The **balance** score (balanced accuracy) averages coverage and the share of reference negatives correctly left without intervention. Higher relevance, coverage, effectiveness, and balance and lower intrusion are preferred. The **frequency** metric measures interventions per reference-positive event: $(TP + FP)/(TP + FN)$, with counts pooled across streams within each scope. A value of $1\times$ indicates equal counts; larger and smaller values indicate more and fewer interventions, respectively. This ratio describes relative intervention volume and is interpreted alongside relevance, coverage, and intrusion.

Implementation details. The primary model is gpt-5.6-luna, with temperature zero and thinking disabled; ProactMem uses the official OpenAI API. Methods share task semantics and the intervention controller policy, with interface-specific prompts and budgets. The event-only, event and trajectory memory, and ProactMem variants share their decision prompt, schema, and context bound of 72 hours and at most 100 events, including the current event. A-MEM and Graphiti augment that bound with their representations. Appendix B.4 lists detailed configurations.

5.2 Main results

We compare proactive-service and memory-based agents on ProactiveStream using the same Luna backbone and intervention controller policy.

Comparisons with external methods. ProactMem achieves higher effectiveness and balance than all evaluated external methods in every scope

(Table 2, RQ1).

Reference sensitivity. Rescoring fixed ProactMem outputs from a separate 450-target diagnostic against the separate-team references yields similar effectiveness on Sports and lower effectiveness on Home and Code relative to the earlier references (Section 4.2). Absolute scores are label-sensitive, especially on Home and Code.

Episode-level analysis. A model-assisted post-hoc analysis on those targets reconstructs goal-and-object episodes from positive event anchors. First timely matching assistance covers an episode; repetition without actionable new evidence is redundant. Episode coverage is highest on Home and lowest on Code; the share of redundant interventions is highest on Home and lowest on Sports. Episode-level analysis complements event-level evaluation by measuring opportunity coverage and repeated interventions. Appendix D retains reference-sensitivity scores and episode-matching records.

5.3 Ablation study and memory analysis

Proactive agent ablation. The event-only variant supplies current and bounded recent events; the event and trajectory memory variant adds source-checked temporal trajectories; ProactMem adds behavioral regularities. These three variants in Figure 3 share Luna, decision prompt, schema, event bound, and controller. The controller uses representation-dependent keys, so the comparison measures memory-layer contributions within the implemented pipeline. The temporal trajectories recover more opportunities, most on Code, with unchanged false-positive counts on Sports and Home and more false positives on Code. The behavioral regularities further improve effectiveness in every scope, recovering more opportunities with fewer false interventions (RQ2). Trajectory abstraction

Table 2: Full-scope results (Luna). The frequency metric is reported in multiples ($\times$); all other scores are percentages. Metric definitions are relevance: share of interventions matching opportunities; coverage: share of opportunities recovered; effectiveness: harmonic mean of relevance and coverage (F1); intrusion: share of reference negatives receiving interventions; balance: average of coverage and correct-silence rate. Bold and underlined values indicate the best and second-best directional scores per scope, including ties.

scope	method	frequency ($\times$)	relevance $\uparrow$	coverage $\uparrow$	effectiveness $\uparrow$	intrusion $\downarrow$	balance $\uparrow$
Sports	ProactiveAgent	2.75	30.0	<u>82.5</u>	43.9	64.5	<u>59.0</u>
	ContextAgent	2.16	<u>31.0</u>	<u>67.1</u>	42.4	<u>50.0</u>	<u>58.6</u>
	Mem0	3.40	28.8	97.8	<u>44.5</u>	81.0	58.4
	A-MEM	2.41	30.6	74.0	43.3	56.0	<u>59.0</u>
	Graphiti	2.20	29.3	64.4	40.3	52.0	56.2
	ProactMem	2.15	35.8	76.7	48.8	46.1	65.3
Home	ProactiveAgent	0.78	34.2	26.8	30.1	6.8	60.0
	ContextAgent	1.38	<u>35.9</u>	49.5	41.6	11.7	68.9
	Mem0	0.73	23.9	17.4	20.1	<u>7.3</u>	55.0
	A-MEM	1.63	34.6	<u>56.3</u>	<u>42.9</u>	14.1	<u>71.1</u>
	Graphiti	1.53	33.4	51.1	40.4	13.5	68.8
	ProactMem	1.52	48.6	73.7	58.6	10.3	81.7
Code	ProactiveAgent	4.45	17.5	<u>78.2</u>	28.7	49.7	64.2
	ContextAgent	2.68	22.5	60.3	32.8	28.2	66.1
	Mem0	0.65	<u>39.8</u>	25.9	31.4	5.3	60.3
	A-MEM	2.55	30.4	77.6	<u>43.7</u>	24.0	<u>76.8</u>
	Graphiti	2.36	26.8	63.2	37.7	23.3	69.9
	ProactMem	2.24	40.1	89.7	55.4	<u>18.1</u>	85.8

expands context; supported, currently applicable regularities improve selectivity.

Validation mechanisms. Table 3 fixes model, prompt, event bound, context capacity, and controller. *w/o replay gate* bypasses activation thresholds for well-formed, source-linked, online-reconstructable candidates, retaining current-applicability matching. *w/o trajectory validation* stores schema-valid, same-entity trajectory extractions without fact checking, retaining weekly construction and replay. Both removals lower relevance and effectiveness (RQ3). Replay gating contributes more effectiveness and improves coverage in every scope. On Code, trajectory validation yields fewer false interventions while retaining the same true-positive count, supporting improved selectivity.

Intervention control. The before suppression diagnostic scores recorded Code proposals while holding later states and decisions fixed. Filtering removes more false-positive than true-positive proposals, yielding higher relevance and effectiveness alongside lower coverage. Appendix C reports expanded ablation and intervention control scores.

5.4 Model scaling

Within-series model scaling. Construction and decision LLMs change together. Table 4 keeps the benchmark, architecture, role-specific prompts, decoding, validation thresholds, and controller policy fixed. The comparison shows increasing effectiveness and decreasing intrusion from GPT-5.6 Luna through GPT-5.6 Terra to GPT-5.6 Sol in every scope. These gains are consistent with stronger abilities to extract task-relevant details, induce conditional behavioral regularities, and perform compositional reasoning over temporal evidence, supporting more accurate memory construction and more selective intervention. Our hierarchical memory architecture engages these abilities in both historical representation and current decision-making. Controlled ablations retain GPT-5.6 Luna.

Cross-family event-only comparison. We compare GPT-5.6 Luna, Gemini 3.8 Flash, DeepSeek V4 Flash, and GLM 5.3 Flash. Task instructions, the context bound of 72 hours and at most 100 events, and the intervention controller policy are shared. Figure 4 pools confusion counts across domains. Luna leads in F1 and recall; Gemini leads in precision.

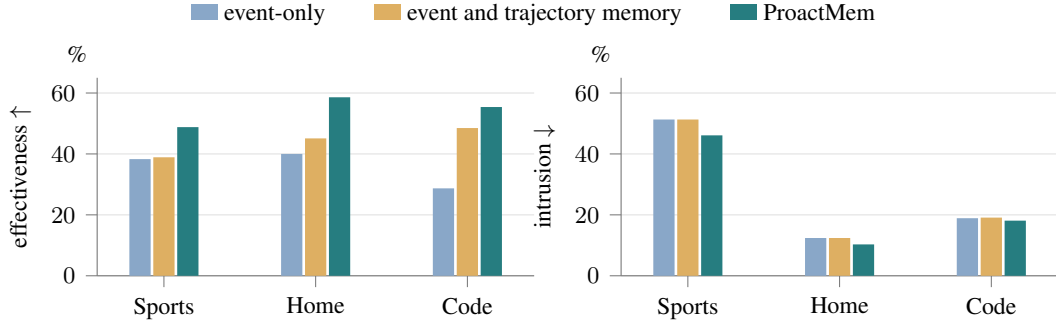


Figure 3: Proactive agent ablation (GPT-5.6 Luna). The event-only, event and trajectory memory, and ProactMem variants share the decision prompt, schema, event bound, and controller policy, applied to each variant’s memory context. The panels report effectiveness and intrusion.

Table 3: Validation ablations and Code intervention control diagnostic (% , Luna). Higher is better. The before suppression diagnostic scores recorded proposals with later states fixed. Bold and underlined values indicate the best and second-best scores per column.

method	Sports		Home		Code	
	relevance	effectiveness	relevance	effectiveness	relevance	effectiveness
w/o replay gate	31.1	42.7	43.6	54.4	33.8	48.1
w/o trajectory validation	<u>34.4</u>	<u>47.5</u>	<u>45.9</u>	<u>56.7</u>	36.1	51.5
before suppression	—	—	—	—	<u>36.7</u>	<u>53.6</u>
ProactMem	35.8	48.8	48.6	58.6	40.1	55.4

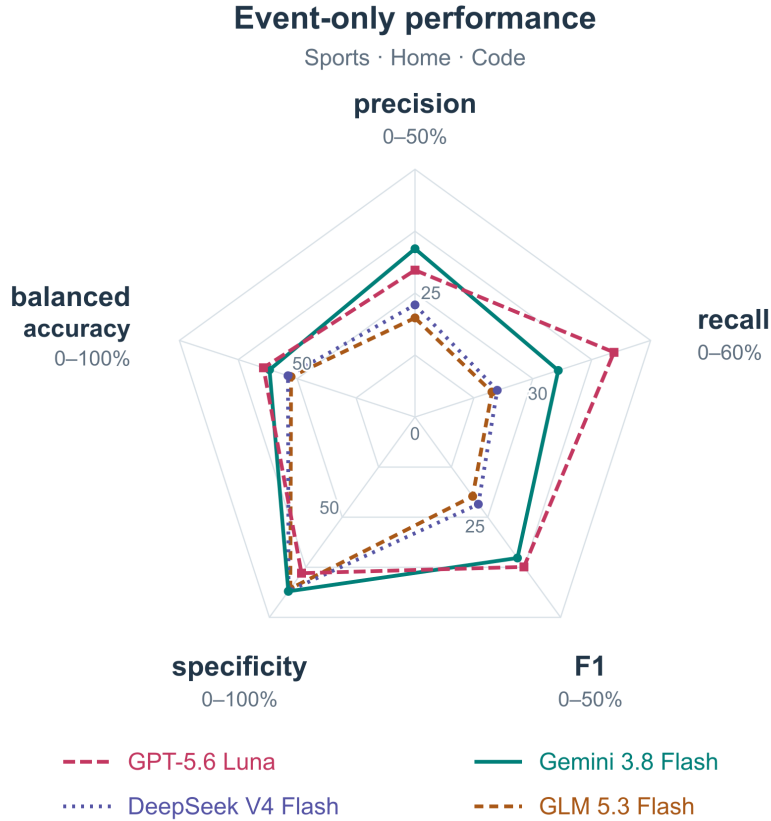


Figure 4: Single-run event-only model comparison on three selected streams. Scores pool event-level confusion counts across domains; specificity equals $1 - \text{intrusion}$. All axes start at zero, with metric-specific ranges.

Table 4: Model scaling (%). Bold and underlined values indicate the best and second-best scores in each column.

model	Sports		Home		Code	
	effectiveness ↑	intrusion ↓	effectiveness ↑	intrusion ↓	effectiveness ↑	intrusion ↓
GPT-5.6 Luna	48.8	46.1	58.6	10.3	55.4	18.1
GPT-5.6 Terra	<u>54.1</u>	<u>39.4</u>	<u>64.4</u>	<u>9.1</u>	<u>60.3</u>	<u>14.9</u>
GPT-5.6 Sol	58.6	34.4	69.3	7.5	66.4	11.8

6 Related work

Agent memory and long-horizon retrieval.

Generative Agents, MemoryBank, and MemGPT organize observations and experience through reflection, updating, or tiered context management (Park et al., 2023; Zhong et al., 2024; Packer et al., 2023). Mem0, A-MEM, and HippoRAG develop extracted memories, linked records, and graph-based retrieval (Chhikara et al., 2025; Xu et al., 2025; Gutiérrez et al., 2024). Graphiti maintains event-derived relations with temporal validity and provenance for retrieval (Zep, 2026). ProactMem derives recurring conditional regularities from temporal trajectories, assesses their historical support, and matches applicable regularities to incoming events.

Proactive decision and memory. ProactiveAgent studies whether an agent should propose a task or remain silent and uses annotated event data to elicit this capability (Lu et al., 2025). ContextAgent combines sensory contexts with historical personas to anticipate service needs (Yang et al., 2025), while PASK couples streaming demand detection with hybrid long-term memory (Xie et al., 2026). Memory-oriented proactivity also supports internal reasoning and execution: MemCog integrates proactive memory exploration into conversational reasoning (Li et al., 2026), and Remember When It Matters uses a separate memory agent to inject selected reminders into long-horizon action agents (Wu et al., 2026). ProactMem focuses on constructing service-relevant temporal memory through source checking, historical replay, and current-event matching.

Evaluation of proactive service. LoCoMo and LongMemEval assess long-term conversational memory and reasoning (Maharana et al., 2024; Wu et al., 2025). Proactive benchmarks examine task proposals, latent-need detection, or memory triggering (Lu et al., 2025; Xie et al., 2026; Li et al., 2026). ProactiveStream evaluates sequential *in-*

tervene or *abstain* decisions as agents accumulate memory within isolated, causally replayed event streams, using episode-aware references to assess intervention timing. Research on notification timing further shows that interruption cost depends on task phase and receptivity (Adamczyk and Bailey, 2004; Iqbal and Bailey, 2007, 2008; Mehrotra et al., 2016). This motivates evaluating opportunity recovery together with unnecessary intervention.

7 Conclusion

We presented ProactMem, a proactive memory framework that combines bounded recent events, source-checked temporal trajectories, and replay-assessed behavioral regularities. Across the three scopes of ProactiveStream, ProactMem achieves higher event-level effectiveness and balance than the evaluated external baselines. Ablations support the contributions of hierarchical organization, trajectory fact validation, and historical replay gating. These findings highlight the value of matching source-grounded, historically supported memory to current observations for service-opportunity detection.

Limitations

ProactiveStream covers low-risk service settings in Sports, Home, and Code under a protocol fixed after joint method and benchmark development. Independent held-out evaluation and cross-family model transfer remain future work. Reference labels represent service opportunities under the annotation protocol; the 450-target review assesses three selected streams. The Code suppression diagnostic holds later states fixed; controller-free behavior requires a sequential comparison. Historical replay assesses observed support; prospective validity under changing behavior requires further study. Our offline metrics evaluate opportunity detection. Recipient benefit, interruption burden, end-to-end cost, and deployment safety require prospective measurements. The reported results support comparisons within the evaluated proto-

cols, with broader generalization depending on additional populations, observation conditions, and model families.

Ethical considerations

Our evaluation is limited to optional activity support, gentle check-ins, and low-risk repository assistance. Medical and high-risk safety decisions require separate domain-specific validation and safeguards. Inference is isolated by stream, and reference labels, episode annotations, and scoring metadata are excluded from model inputs.

Longitudinal events and derived memories can reveal sensitive routines even when identifiers are pseudonymized. Event content, prediction traces, and inferred trajectories require access and retention protections; external-model use should minimize transmitted information and respect source-data permissions.

Deployment should require informed opt-in and allow inspection, correction, deletion, retention control, and disabling proactive assistance. An intervention prediction represents a service proposal; consequential action requires separate authorization. These deployment safeguards require separate implementation and evaluation.

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A Benchmark and annotation protocol

A.1 Public events and availability

Sports. HeartSteps mobile activity contexts (Klasnja et al., 2019) supply 1,456 meta-events from nine complete participant streams over 367 observed participant-days. Public fields include decision-time availability, recognized activity, recent steps, location category, weather, and local decision slot. Randomized treatment assignments and future behavioral outcomes are excluded; services concern optional activity support.

Home. CASAS ambient-sensor traces (Cook et al., 2013) supply four homes with 14 consecutive days each. Per local day, ten-minute source blocks are assigned exactly once to 28 regular 50-minute segments and one 40-minute tail, yielding 406 events per home. A segment becomes available only after interval closure and contains within-interval motion counts, room sequences, door actions, and compact numeric summaries. Gentle check-ins exclude disease, fall, travel, danger, and unsupported preference inference.

Code. Public repository activity, including GH Archive records (Grigorik, 2026), supplies 1,459 events across four repositories with 174 reference interventions. Pushes, issue updates, pull-request actions, and reviews retain GitHub’s native event granularity, with one decision per event. Records are ordered by timestamp and then event ID. Each repository owns isolated memory and intervention state even when contributors overlap. Services concern low-risk triage, review, and maintenance; a native PushEvent is an observation, not an agent intervention.

Public-input contract. Both label classes share the same scope-specific schema: event and stream identifiers, timestamps, and contemporaneously observable fields. Reference labels, candidate flags, episode annotations, memory-level markers, and scoring masks remain evaluator-private. Evaluation includes early events before temporal trajectories or behavioral regularities are available.

A.2 Reference labels and causal replay

Annotators apply the opportunity and novelty semantics in Section 4.2 to each current event and its complete prior same-stream history, independently of a tested method’s memory. LLM-assisted

screening prioritizes the review queue; three team members independently assign labels while blinded to evaluated-system decisions, followed by cross-review and adjudication of every public event.

The Home aggregation rule precedes event construction. Each evaluation version records the event inventory, labels, intervention prompt, event memory bound, and validation settings; these are frozen before the corresponding main runs. Evaluation follows a fixed protocol on the jointly developed benchmark.

Every stream starts from empty method state and is replayed chronologically under the causal execution order in Section 3.3. Current events are committed only after the output is finalized; maintenance incorporating an event affects later decisions. Labels are joined offline, and reference episode states and scoring metadata remain organizer-private.

B Implementation and experimental configuration

B.1 trajectory evidence and replay

The trajectory construction step uses an independent buffer of all committed events from the open local day, cleared after processing. Activity segments contain a local interval, typed observed, context-before, and outcome fields, plus source-event identifiers. The checks in Equation 5 canonicalize scalar values, accept exact or schema-preserving suffix field paths, and require consistent entities, intervals, and resolved citations. Validation and replay operate on typed evidence. Accepted trajectories remain in the current construction cycle after their events leave event memory.

A cycle contains seven completed daily records, each assigned to one cycle, and may span more than seven calendar days. Empty or unobservable records also advance the cycle. Consolidation resets the online trajectory view while preserving records for audit. Replay follows Equation 6: a triggered, observable day is fulfilled when the expected behavior occurs at or after the trigger within the window; otherwise it is violated. Missing observations are censored, each date counts at most once, and the base rate includes triggered and other observable days under the same time and weekday restrictions. Confidence is zero with no eligible days; lift is zero with zero base rate.

B.2 Lifecycle and current matching

Lifecycle eligibility is separate from occurrence state and service-episode state. After admission, confidence combines replay counts and subsequent outcomes in a support ratio smoothed with one initial success and one initial failure; censored observations preserve both counts. Two consecutive violations move an active regularity to *weakening* with lower retrieval priority; four consecutive violations move it to *dormant*. A fulfilled observation restores a weakening regularity to active when confidence is at least 0.5. Dormant regularities must return to the candidate state and pass replay readmission.

Current matching. Current matching is deterministic. Retrieval admits upcoming, pending, or deviated occurrences of active or weakening regularities. Occurrence priority is deviated, followed by pending and upcoming.

B.3 Intervention control

In Sports and Home, the intervention key combines scope-specific episode state, proposed service action and target, and sorted regularity identifiers available in context. Plain summaries or graph facts supply an empty regularity-identifier list. Sports state includes availability and activity; Home state includes activity and room and onset, confirmed, and sustained phases at one, two, and three or more consecutive matching observations. Without an explicit episode key, scope-specific time or target fields identify the opportunity, with unchanged evidence additionally required for duplicate suppression. State changes, sequence restarts, service or regularity identifier changes, and local-day transitions can yield new keys.

Code uses repository-object identity and the most recent allowed intervention’s mode, recipient, content, and firing condition. An exact match converts a repeated create proposal to *abstain*; changed signatures and other episode operations pass this check. Representation-dependent key evidence can yield different suppression decisions across memory interfaces.

B.4 Model and method interfaces

Decision and memory-construction calls use the model, API, and decoding settings in Section 5.1, with separate output budgets. Decision limits are interface-specific (256 or 512 tokens); construction limits are configured by call role. Table 5 con-

solidates the external-method interfaces. Methods retain their own prompts, retrieval, and computational budgets.

Graphiti supplies relation facts directly, using the task policy and output contract with memory-specific guidance; its ingestion episodes correspond to source records. The cross-family event-only comparison uses backend-specific output-format and recovery settings.

Scoring. Using final-decision confusion counts, relevance is $TP/(TP + FP)$, coverage is $TP/(TP + FN)$, effectiveness is $2TP/(2TP + FP + FN)$, intrusion is $FP/(FP + TN)$, and balance is $\frac{1}{2}[TP/(TP + FN) + TN/(TN + FP)]$.

Table 5: External-method memory interfaces and implementation settings.

method	interface and settings
ProactiveAgent	Released evaluation prompt and proposal/null contract, adapted to the study backbone; causal event–decision history capped at 1,000 events including the current event per request (Lu et al., 2025; THUNLP, 2026).
ContextAgent	In-context proactive-service formulation adapted to public event representations and the study backbone (Yang et al., 2025).
Mem0	Open-source mem0ai==2.0.12 extraction, consolidation, and retrieval; local all-MiniLM-L6-v2 embeddings; top-8 same-entity retrieval with search threshold 0.1 (Chhikara et al., 2025; Mem0, 2026).
A-MEM	Linked notes with association updates; retrieved notes augment the bounded event context (Xu et al., 2025).
Graphiti	Graphiti 0.30.2 with Neo4j and local 384-dimensional all-MiniLM-L6-v2 embeddings; committed events enter a stream-isolated graph. The current event queries native hybrid edge search with reciprocal-rank fusion, returning at most ten relation facts with validity fields and source-record identifiers (Zep, 2026).

C Additional ablation and intervention control results

Table 6 supplements the relevance and effectiveness scores in Table 3 with frequency and coverage. On Code, suppression removes 66 false-positive and 17 true-positive proposals; coverage falls from 99.4% to 89.7%. The diagnostic scores recorded proposals while holding later states fixed. For the variants in Figure 3, temporal trajectories yield aggregate net gains of 4, 15, and 64 true positives on Sports, Home, and Code, respectively; behavioral regularities yield further net gains of 57, 33, and 22.

D Reference and episode diagnostics

D.1 Reference sensitivity

The diagnostic selects 150 consecutive targets from one stream per scope using a fixed SHA256 ranking of eligible streams and contiguous starts based only on public metadata, with at least seven days of preceding history. The streams are Sports participant_35, Home home_0004, and Code repository callstack/react-native-pager-view. Their target periods are December 16, 2015–January 18, 2016; June 23–28, 2011; and June 21–November 4, 2025. All 37, 203, and 152 preceding events are processed chronologically, respectively: 842 public events supply context for 450 scored targets.

A separate three-reviewer team independently labels each target with its prior same-stream history while blinded to evaluated-system outputs, then discusses judgments to obtain consensus references. Relative to the earlier diagnostic references, Sports and Home each have 13 changes from *intervene* to *abstain*, with 4 and 0 changes in the reverse

Table 6: Additional evidence-check ablation and Code intervention control metrics. The frequency metric is reported in multiples ($\times$) and coverage in percentages. Bold and underlined values indicate the best and second-best coverage per scope, including ties. Scores for relevance and effectiveness appear in Table 3.

scope	method	frequency ($\times$)	coverage $\uparrow$
Sports	w/o replay gate	2.19	<u>68.2</u>
	w/o trajectory validation	2.23	76.7
	ProactMem	2.15	76.7
Home	w/o replay gate	1.65	72.1
	w/o trajectory validation	1.62	74.2
	ProactMem	1.52	<u>73.7</u>
Code	w/o replay gate	2.47	83.3
	w/o trajectory validation	2.48	<u>89.7</u>
	before suppression	2.71	99.4
	ProactMem	2.24	<u>89.7</u>

direction, respectively; Code has 6 removals and 16 additions. These counts describe differences between the earlier diagnostic references and the separate team’s judgments on 450 targets across three selected streams.

Table 7 scores identical recorded ProactMem outputs against these two sets of references. The predictions come from separate diagnostic runs on the selected targets. The diagnostic uses Luna with temperature zero, thinking disabled, decision limits of 256 tokens for Sports and Home and 512 for Code, and the same context bound of 72 hours and at most 100 events, trajectory validation, replay gating, and intervention control policy. Immediate and scheduled services both count as *intervene*.

Table 7: Reference sensitivity on 150 targets per scope. I is the number of final interventions; positives and confusion counts use the separate team’s references. Effectiveness is F1 (%). Earlier diagnostic references use original Sports/Home labels and 122 original plus 28 additional Code annotations. Both columns score the same diagnostic outputs.

Scope	Positives	I	TP	FP	FN	TN	Earlier diagnostic Effectiveness	Separate-team Effectiveness
Sports	27	86	24	62	3	61	42.6	42.5
Home	15	39	15	24	0	111	68.7	55.6
Code	34	42	18	24	16	92	57.6	47.4

Table 8: ProactMem episode diagnostic on 150 targets per scope. Covered/New includes Episode Coverage (%) in parentheses; SR and RIR are percentages. I is the sum of first valid (F), valuable follow-up (V), redundant (R), uncertain (U), unmatched, late, and unresolved outcomes. $U = 0$ in all three scopes; unresolved semantic or condition matches remain separate from supported or redundant assistance.

Scope	Covered/New (Coverage)	I	F	V	R	Unmatched	Late	Unresolved	SR	RIR
Sports	23/26 (88.5)	86	23	1	1	60	0	1	27.9	1.2
Home	10/10 (100.0)	39	10	6	17	6	0	0	41.0	43.6
Code	14/31 (45.2)	42	14	3	4	16	4	1	40.5	9.5

D.2 Episode references and matching

Each reference specifies a service goal and object, causal evidence, onset, explicitly listed valid observations, and any deadline or closure evidence. Semantic matching uses these fields to associate interventions with reference episodes.

Matching and follow-up. Episode coverage follows ProactMem’s own interventions, initialized from recorded prefix outputs; fixed event-label novelty instead follows the reference sequence (Equation 9). Only target interventions are scored. An intervention must match the goal and object at a listed valid observation. Its proposed service time is the decision time plus its specified delay (zero for immediate service), and must meet any explicit episode deadline. The first timely match covers the episode. Each output receives at most one credit: the first uncovered compatible episode in the fixed reference order, otherwise the first already covered one.

Subsequent matches are compared with the method’s preceding assistance for that episode. A valuable follow-up requires new causal evidence that changes the actionable service or next step, and a message addressing that change. Rephrasing or continued status alone is redundant; value requiring unobserved delivery, recipient response, or completion remains uncertain. Valuable follow-ups do not create additional covered episodes.

Metrics and boundaries. Let I count all target interventions, F first valid interventions, V valuable follow-ups, R redundant follow-ups, and U uncertain follow-ups. Episode coverage divides covered new episodes by all new reference episodes; supported relevance is $SR = (F + V)/I$, and redundant intervention rate is $RIR = R/I$. Uncertain, unmatched, late, and unresolved outputs remain in I but receive no supported-assistance credit. Table 8 combines the rates and outcome counts.

Episode coverage is conditional on the observed segment and onset convention. The main diagnostic treats the first Home opportunity as target-onset; treating it as a continuation from the prefix instead covers all 9 new opportunities. These exploratory scores depend on the constructed references and semantic matches.